Galaba Vamsi

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SUMMARY

I build evaluation and experiment infrastructure for AI agents and robot policies, and I publish what did not work. Co-author of Inspect Robots (CoRL 2026 workshop); 15 merged PRs at 3 YC startups; Top 1% in JEE Advanced.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 – May 2027
Coursework: AI and ML, Deep Learning for Computer Vision, Optimal Control, Data Structures and Algorithms

RESEARCH AND EXPERIENCE

Co-founder, Robotropic (physical-AI startup) | robotropic.com Oct 2026 – Present
- Building Harness to roll out new robot policies (VLA checkpoints) cell by cell, with shadow-run scorecards and rollback.
- Lead its change gateway, which checks every human or AI-agent action against the cell's safety limits before it runs.

Research Intern, IIT Bhilai | Advisor: Dr. Arzad Alam Kherani May 2025 – Jul 2025
- Built RIS-SIM, an open-source Python emulator for smart radio surfaces (RIS); co-author, IEEE INDICON 2025.

EVALUATION AND EXPERIMENT INFRASTRUCTURE (OPEN SOURCE)

Inspect Robots, Robocurve (YC S26): paper co-author; 2 merged PRs, 3 in review Sep 2026 – Oct 2026
Evals for LLM and VLA robot policies | Python | github.com/robocurve/inspect-robots
- In review: checkpoint/resume for long robot evals; safety aborts and hardware faults never retried; 80 tests (#514).
- Merged: configurable LLM-API retries honoring Retry-After across 5 wire protocols, 22 test cases (#476).
- Published inspect-robots-wandb on PyPI: Weights & Biases logging for robot-policy evals, 100% test coverage.

HFlow, Hebbian Robotics (YC S26): 8 merged PRs, 7 issues filed (all fixed) Sep 2026 – Oct 2026
Robot training-data SDK | Python, Apache Arrow, MCAP, LeRobot | github.com/Hebbian-Robotics/hflow
- Fixed robot-log export silently dropping fields missing from the first message (#657); stopped 12-bit depth loss (#400).
- Built a reproducible benchmark (900 decoded 1080p frames per check); no safe speedup, so the code stayed as is (#367).

PROJECTS

World Model for Robot Manipulation: JEPA on LeRobot Push-T Aug 2026
PyTorch | write-up: galabavamsi.com/blog
- Trained a 1.9M-parameter action-conditioned world model on 160 Push-T episodes in 7 minutes on an 8 GB GPU.
- Its latents captured the block's state, but rollouts drifted after 2 steps and the planner scored 0; published the analysis.

Pehchaan: voice-clone fraud detection (team of 2; Top 8 of 4,500+ teams) Oct 2026
Python, PyTorch, FastAPI, Kotlin | github.com/Galabavamsi/pehchaan
- Fine-tuned XLS-R + AASIST to 4.8% EER (34.3% off the shelf); on 41 real calls, error fell from 24.6% to 6.8%.

Urumi MCP Gateway: one AI-tool (MCP) endpoint for a whole team Oct 2026
TypeScript, PostgreSQL, Kubernetes (k3s), Helm, AWS | github.com/Galabavamsi/urumi-mcp-gateway
- Per-user API keys, role-based tool permissions, an audit log and human approval before write actions; 321 tests.

sremu: timed radio-control emulator 2026
Python, ZeroMQ, GNU Radio | solo, built with Claude Code | github.com/Galabavamsi/sremu
- On AWS, 99.8%+ of control commands sent 4 ms ahead took effect at their exact radio sample; 2,136 passing tests.

WEAR Dataset Challenge (Kaggle): 5th of 232 teams 2026
- Wearable IMU + video, 4 unseen subjects: private macro-F1 0.93137; caught a leak that looked like a gain.

SKILLS

AI: LLM APIs, AI agents and tool use, MCP, agent and policy evaluation, PyTorch, VLA fine-tuning (π0.5), JEPA
Languages: Python, TypeScript, C, C++, Rust, SQL
Infrastructure: Linux, Docker, Kubernetes, Helm, AWS, GitHub Actions, PostgreSQL, Weights & Biases, Prometheus, pytest

ACHIEVEMENTS

- Top 1% in JEE Advanced (2023); Prime Minister's Scholarship Scheme scholar (2024)
- Top 8 of 4,500+ teams, RAKSHAM AI Cybersecurity Hackathon (2026); 5th of 23 IITs, Inter IIT Tech Meet 14.0 (2025)
- All India Rank 200 (2026) and 747 (2025), Amazon ML Challenge (team)

PUBLICATIONS

- Co-author, Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI, arXiv:2610.06306; CoRL 2026 workshop
- Co-author, 2 RIS papers: IEEE INDICON 2025 and *Ad Hoc Networks* (Elsevier, 2026)
- Sole author, *Timed RIS Control in the Loop* (sremu), draft in preparation for arXiv